

Tarea 6.4

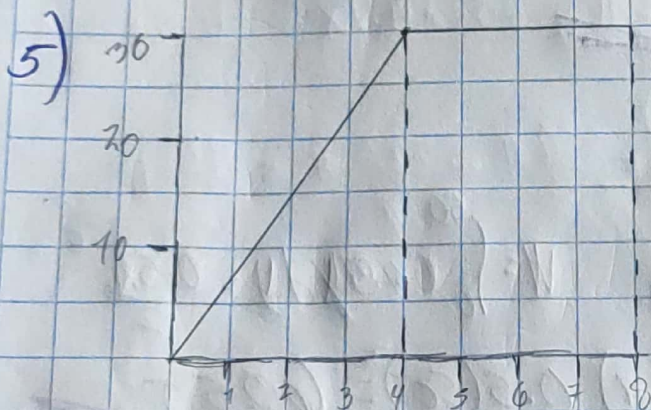
$$3) F = \frac{10}{(1+x)^2} \text{ libras}$$

$$\text{dist} = 9 \text{ pies}$$

$$m = 1+x \\ dm = dx$$

$$W = \int_0^9 \frac{10}{(1+x)^2} dx = \int_0^9 \frac{10}{m^2} dm = 10 \int_0^9 \frac{1}{m^2} dm = 10 \left(-\frac{1}{m} \right) = 10 \left(-\frac{1}{1+x} \right) \Big|_0^9$$

$$10 \left(-\frac{1}{1+9} \right) = \underline{9 \text{ lb/pies}}$$



$$\Delta = \frac{b+h}{2} = \frac{4+30}{2} = 60$$

$$\square = b \Delta h = 4 \cdot 30 = 120 + 180 \text{ J}$$

$$a) 42 - 30 = 12 = 0.12 \text{ m}$$

$$W = \int Kx = \frac{1}{2} Kx^2$$

$$2 = \frac{1}{2} Kx^2$$

$$4 = Kx^2$$

$$4 = K(0.12)^2$$

$$4 = K(0.0144)$$

$$K = \frac{2500}{9} \text{ N/m}$$

$$a) W = \int_{0.05}^{0.10} \frac{2500}{9} x dx = \frac{2500}{18} x^2 \Big|_{0.05}^{0.10} = \frac{25}{18} - \frac{25}{72} = \underline{\underline{\frac{25}{24} \text{ J}}}$$

$$b) F = Kx$$

$$30 \text{ N} = \left(\frac{2500}{9} \right) x$$

$$x = 0.108 \text{ m} = \underline{\underline{10.8 \text{ cm}}}$$

71) $F = KX$

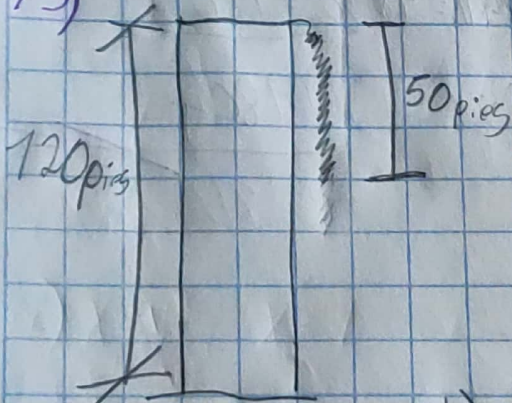
$$W_1 = \int_0^{0.1} KX dx = \left. \frac{KX^2}{2} \right|_0^{0.1} = \frac{K(0.1)^2}{2} = K \frac{1}{200}$$

$$W_2 = \int_{0.1}^{0.2} KX dx = \left. \frac{KX^2}{2} \right|_{0.1}^{0.2} = \left(\frac{K(0.2)^2}{2} \right) - \left(\frac{K(0.1)^2}{2} \right) = \frac{1}{50} - \frac{1}{200} = \frac{3}{200} K$$

$$\frac{W_2}{W_1} = \frac{\frac{3}{200}}{\frac{1}{200}} = 3$$

$$W_2 = 3W_1$$

73)



$$W = \int_0^{50} 0.5x dy = \left. \frac{0.5x^2}{2} \right|_0^{50}$$

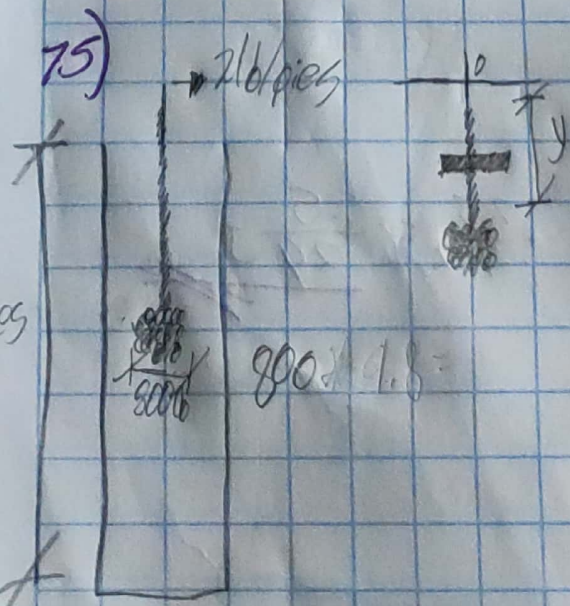
$$a) \frac{0.5(50)^2}{2} = 625$$

$$(0.5(6/pie) \times 1.8$$

$$b) W = \int_{25}^{50} 0.5x dx = \left(\frac{0.5x^2}{2} \right)_{25}^{50}$$

$$W = \left(\frac{0.5(50)^2}{2} \right) - \left(\frac{0.5(25)^2}{2} \right) = \frac{7875}{4}$$

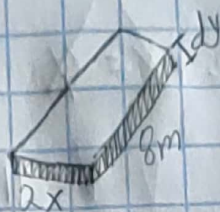
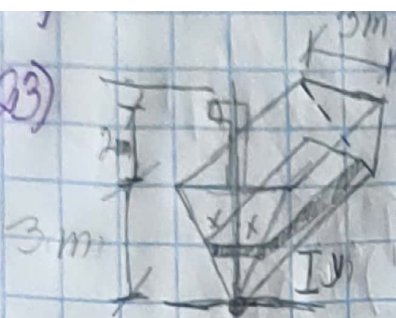
75)



$$W = \int_0^{500} 800 + 2y dy = 800y + y^2 \Big|_0^{500}$$

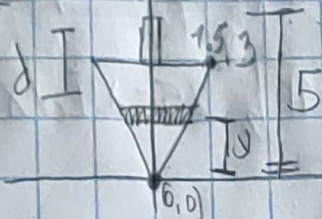
$$800(500) + 500^2 = 650,000 \text{ pies/lb}$$

23)



$$V = bh = (2x)(8)dy$$

$$V = 16x dy$$



$$m = \frac{3-0}{1.5-0} = 2$$

$$y = 2x$$

$$\frac{y}{2} = x$$

$$V = 16x dy$$

$$W = \rho g d V$$

$$\rho g = 9,800 \text{ N/m}^3$$

$$d = 5 - y$$

$$V = 16\left(\frac{y}{2}\right) dy = 8y dy$$

$$1000 \times 9.8$$

$$W = \int_0^3 9,800 (5-y)(8y) dy =$$

$V = \text{area} \times \text{thickness}$

$$W = \rho g d V$$

$$\rho g = 9,800$$

$$d = 5 - y$$

$$V = b h d y$$

$$V = 16x dy = (2x)(8) dy$$

$$V = \left(2\left(\frac{y}{2}\right)\right) 8 dy$$

$$W = \int_0^3 9,800 (5-y)(8y) dy$$

$$V = \left(16\left(\frac{y}{2}\right)\right) dy = 8y dy$$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3-0}{1.5-0} = 2$$

$$y - y_0 = m(x - x_0)$$

$$y - y_0 = m(x - x_0)$$

$$y - y_0 = m(x - x_0)$$

$$y - 0 = 2(x - 0)$$

$$y = 2x$$

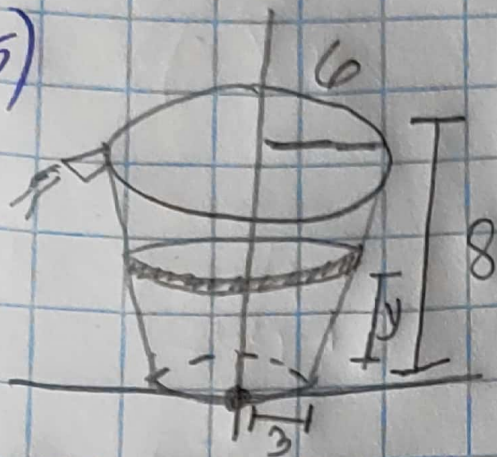
$$y = 2x$$

$$\frac{y}{2} = x$$

$$9,800 \int_0^3 (40y - 8y^2) dy = 9,800 \left(20y^2 - \frac{8}{3}y^3 \right) \Big|_0^3$$

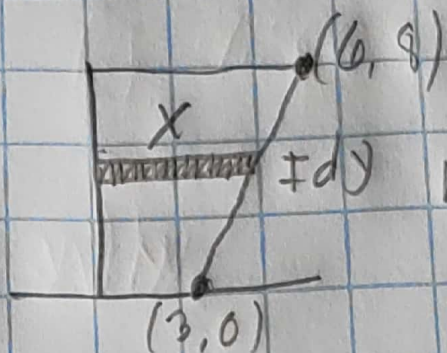
$$W = 152,800 \text{ J}$$

25)



$$V = \pi x^2 dy$$

$$V = \pi \left(\frac{3y}{8}\right)^2 dy$$



$$m = \frac{8-0}{6-0} = \frac{8}{3}$$

$$y - y_0 = m(x - x_0)$$

$$y - 0 = \frac{8}{3}(x - 0)$$

$$y = \frac{8}{3}x$$

$$W = \rho g dV$$

$$\rho g = 62.4 \text{ lb/ft}^3$$

$$d = 8 - y$$

$$V = \pi r^2$$

$$W = \int_0^8 62.4(8-y) \left(\pi \left(\frac{3x}{8}\right)^2\right) dy$$

$$W = \frac{14976}{5} \pi$$